

Applied Data Science Capstone

Falcon 9 First-Stage Landing Prediction

Data Science Capstone Project Report

Prepared from the supplied IBM Skills Network laboratory notebooks. Final PDF for AI-Graded Submission.

Executive Summary

- Objective: analyze historical Falcon 9 launch data and build a binary classifier for first-stage landing success.
- Pipeline: SpaceX API collection → Wikipedia web scraping → data wrangling → EDA/SQL → Folium and Plotly Dash → predictive classification.
- The wrangled modeling dataset contains 90 Falcon 9 launch records; 60 are labeled successful landings, giving an overall success rate of about 66.7%.
- Success becomes much more frequent in later flights and years; launch site and orbit type also show distinct patterns.
- On the documented 18-sample test split: Logistic Regression 83.3%, SVM 83.3%, Decision Tree 94.4%, KNN 88.9%.

Introduction & Problem Statement

- SpaceX reduces launch cost through reuse of the Falcon 9 first stage. The project analyzes whether a first stage lands successfully.
- Target: Class = 1 for a successful landing outcome and Class = 0 otherwise.
- Questions: which sites and orbits show different outcomes, how payload relates to landing, how success changes over time, and whether machine learning can predict the outcome.

Data Collection — SpaceX REST API

- GET requests collect historical launch records.
- Linked identifiers for rockets, payloads, launchpads and cores enrich each record with booster version, payload mass, orbit, launch-site coordinates and core-reuse attributes.
- The final API dataframe is restricted to Falcon 9 launches through the course cutoff date.

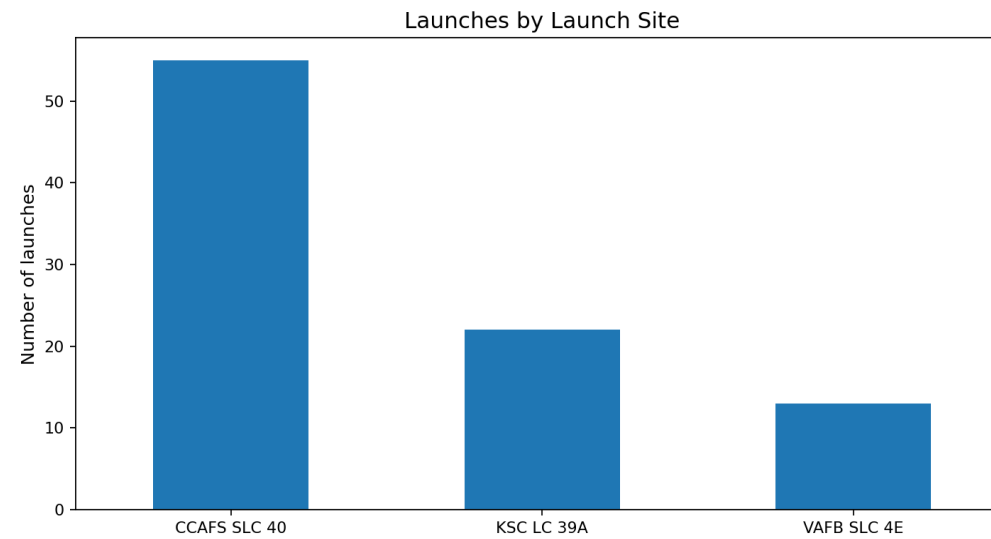
Key fields: FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, Latitude.

Data Collection — Web Scraping

- A fixed Wikipedia Falcon 9 launch-history page is requested with a browser-style User-Agent.
- BeautifulSoup locates launch tables and parses headers and rows.
- Helper functions extract date/time, booster version, landing status and payload mass while handling HTML annotations.
- The parsed records are exported as `spacex_web_scraped.csv`.

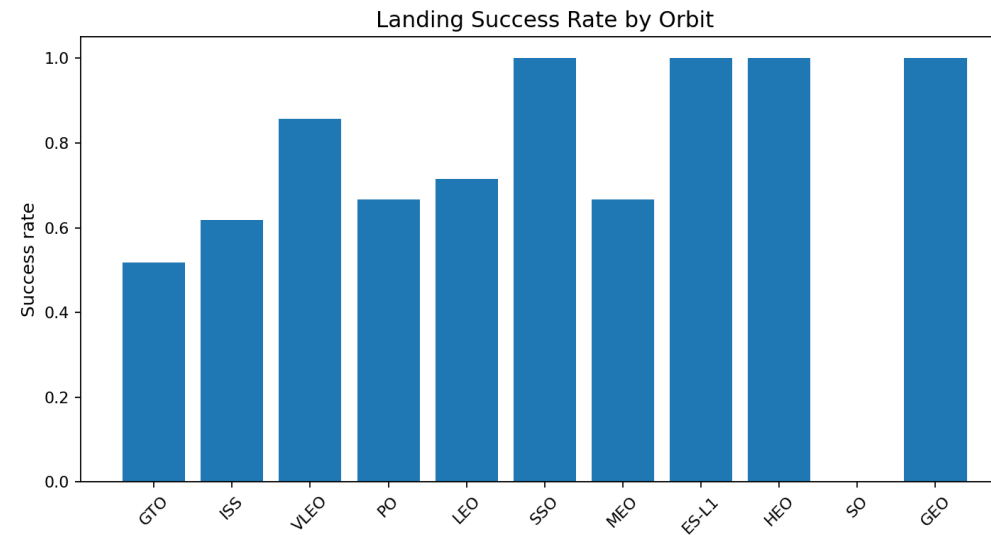
Data Wrangling Methodology

- Filter to Falcon 9 launches and reset FlightNumber after filtering.
- Calculate missing-value statistics and replace missing PayloadMass values with the column mean.
- LandingPad nulls remain meaningful because some missions did not use a landing pad.
- Create the binary Class target from Outcome.



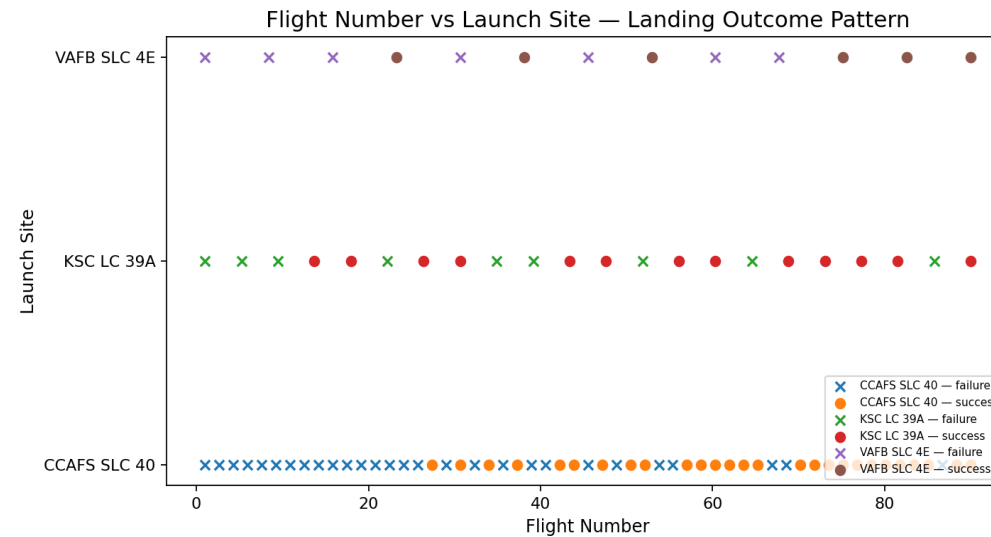
Launch-site counts: CCAFS SLC 40 = 55, KSC LC 39A = 22, VAFB SLC 4E = 13.

EDA with Visualization — Orbit Success



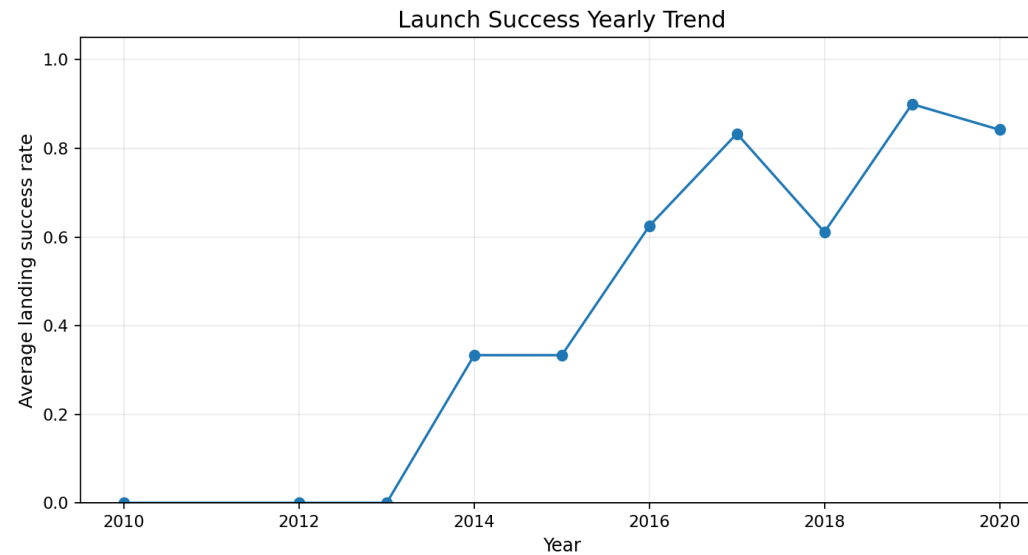
- GTO is the most frequent orbit (27 missions), followed by ISS (21) and VLEO (14).
- ES-L1, GEO, HEO and SSO show 100% success in this sample, but several have only one observation.
- VLEO is about 85.7% successful over 14 missions; GTO is about 51.9%.

EDA with Visualization — Flight Number & Payload



- Later flight numbers contain a much higher concentration of successful landings.
- Payload mass and outcome overlap: payload alone does not explain landing success.
- The visualization supports the course observation that operational experience is strongly associated with improved landing outcomes.

Launch Success Yearly Trend

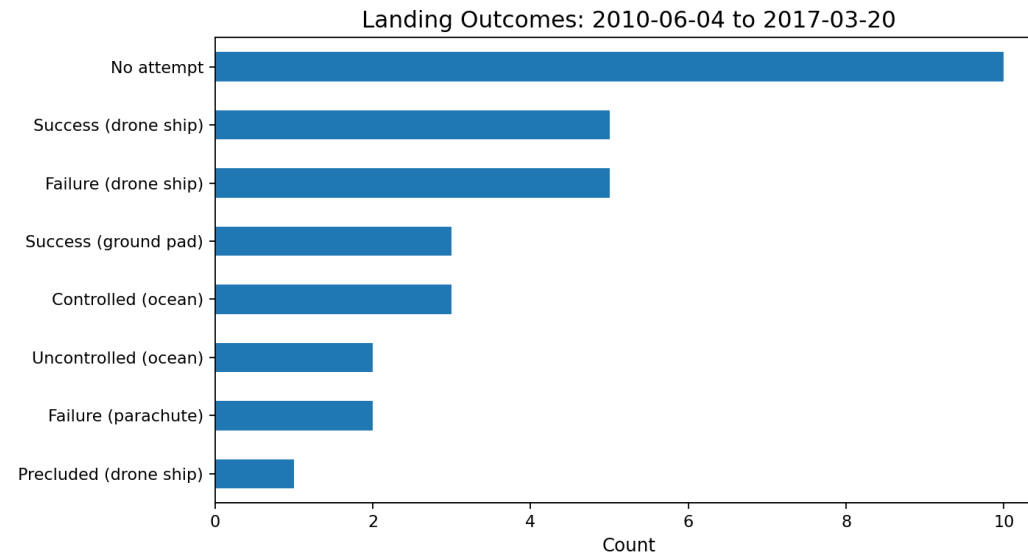


- Sample rates rise from 0% in the earliest observations to materially higher rates in the later reusable-booster period.
- Approximate rates: 2014 33%, 2015 33%, 2016 63%, 2017 83%, 2018 61%, 2019 90%, 2020 84%.
- The trend is descriptive for this course dataset and should not be generalized beyond the sample.

EDA with SQL — Core Results

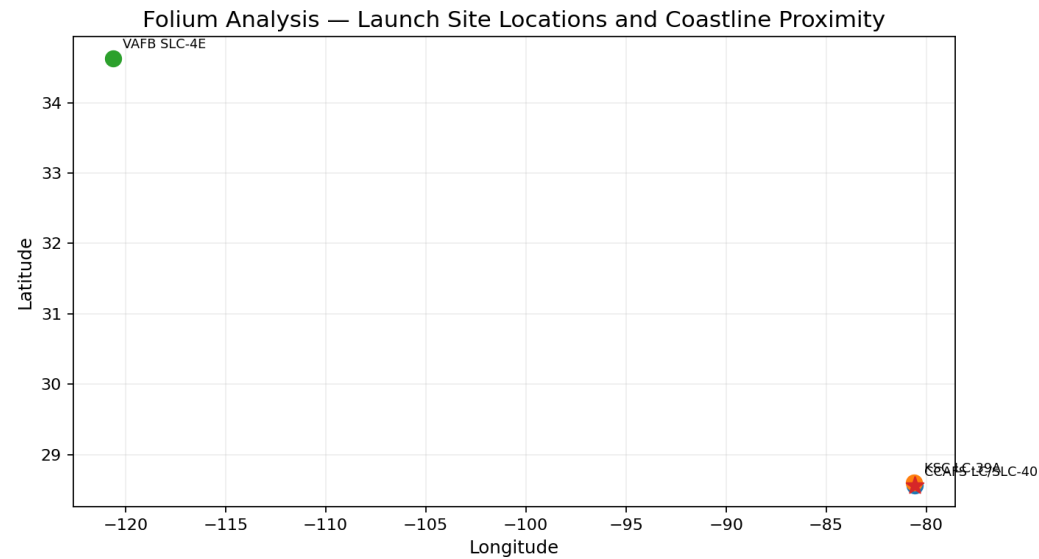
- Unique launch sites: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E.
- Total payload mass for NASA (CRS): 45,596 kg.
- Average payload mass for exact booster version F9 v1.1: 2,928.4 kg.
- First successful ground-pad landing: 2015-12-22.
- 2015 drone-ship failures: F9 v1.1 B1012 and F9 v1.1 B1015 at CCAFS LC-40.

EDA with SQL — Landing Outcome Ranking



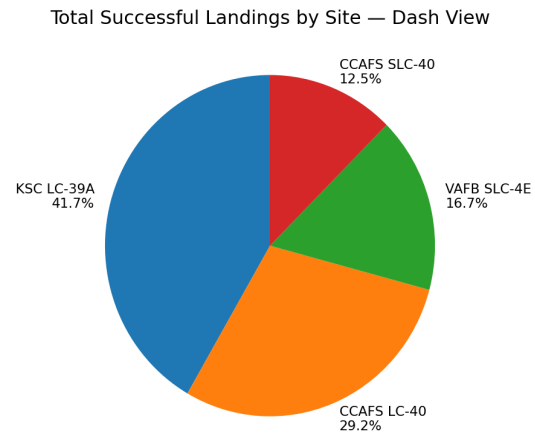
Between 2010-06-04 and 2017-03-20, the ranking is led by 'No attempt' (10), followed by 'Failure (drone ship)' (5) and 'Success (drone ship)' (5).

Interactive Visual Analytics — Folium



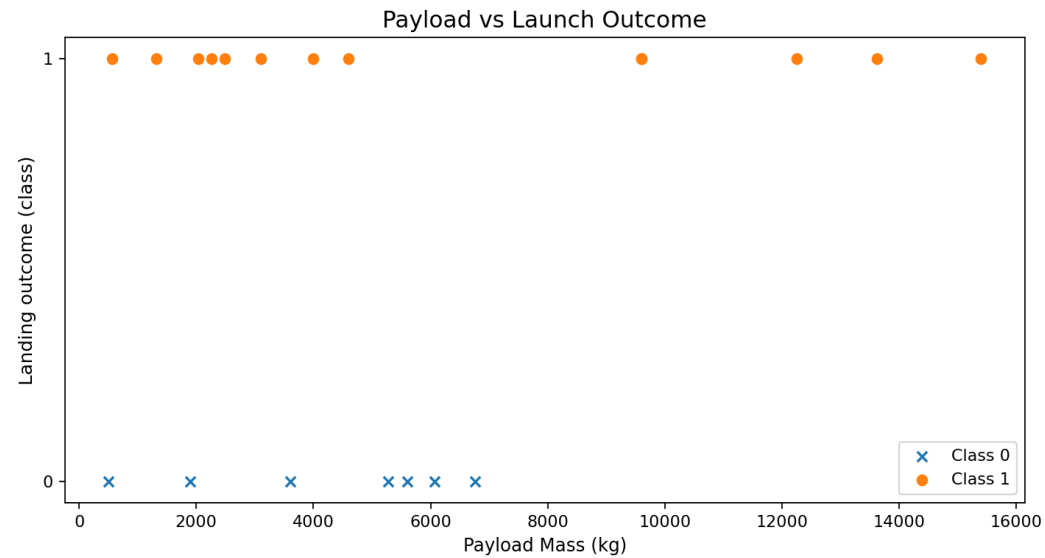
- Launch sites are marked by latitude/longitude and overlapping launch records are clustered.
- Green/red markers distinguish successful and unsuccessful landing records.
- MousePosition and distance calculations support coastline, railway, highway and city proximity analysis.
- A representative Cape Canaveral coastline check is roughly 0.9 km from the selected launch-site coordinate.

Interactive Visual Analytics — Plotly Dash



- Dashboard controls: launch-site dropdown, success pie chart, payload range slider and payload-vs-outcome scatter.
- Historical dashboard labels split successful landings as KSC LC-39A 41.7%, CCAFS LC-40 29.2%, VAFB SLC-4E 16.7% and CCAFS SLC-40 12.5%.
- KSC LC-39A has 10 successful and 3 unsuccessful landing outcomes in the dashboard view (76.9% success).

Plotly Dash — Payload vs Outcome

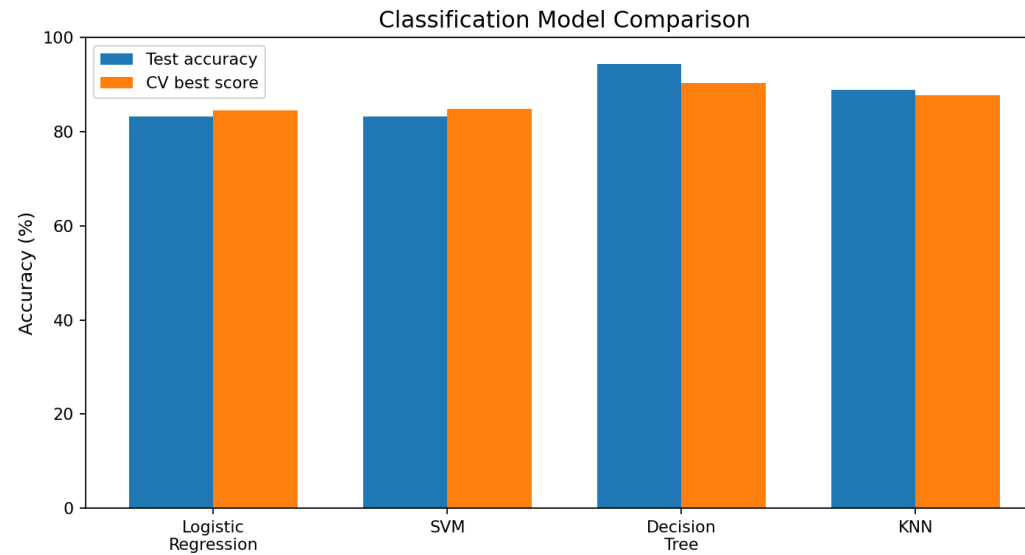


- The payload slider allows focused inspection of light, medium and heavy payload ranges.
- Successful landings appear across a broad payload range, with later booster generations strongly represented in the higher-payload successful region.
- The dashboard is exploratory and describes associations rather than proving causality.

Predictive Analysis — Methodology

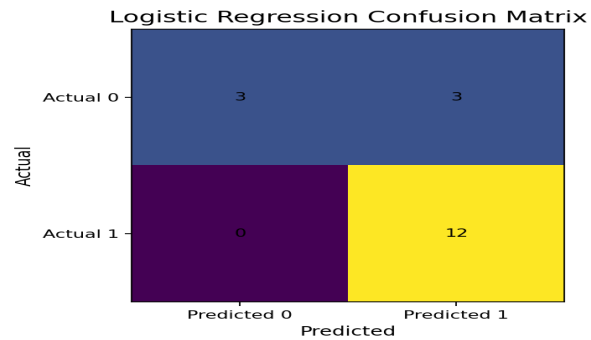
- Y is created from Class.
- X is standardized with StandardScaler.
- 80/20 train-test split with random_state=2 produces 72 training and 18 test observations.
- GridSearchCV with 10-fold cross-validation tunes Logistic Regression, SVM, Decision Tree and KNN.
- Each classifier is evaluated using test accuracy and a confusion matrix.

Predictive Analysis — Model Comparison



Documented test-set accuracy: Logistic Regression 83.3%, SVM 83.3%, Decision Tree 94.4%, KNN 88.9%. Decision Tree's best cross-validation score is about 90.4%.

Predictive Analysis — Confusion Matrix



- Logistic Regression correctly identified 12 successful landings and 3 non-landings in the documented test split.
- The course material notes false positives as the main error pattern for the baseline logistic model.
- Because the test set has only 18 observations, the reported accuracy is best treated as a course-exercise metric.

Conclusion & Key Insights

- The project forms a complete data-science workflow from collection to predictive analysis.
- Landing success increases substantially with later flight numbers and later years in this dataset.
- Launch site and orbit type have distinct success patterns; KSC LC-39A shows 10 successes out of 13 in the dashboard view.
- Payload mass is informative for exploration but is not sufficient on its own to explain landing outcome.
- Decision Tree has the highest documented test accuracy on the 18-sample split; other models provide useful baselines.
- Folium and Dash add spatial and interactive perspectives that complement static EDA and SQL.

Appendix — GitHub & Submission

- Upload the completed notebooks and Dash application to your own GitHub repository.
- Replace the placeholder below with the actual repository URL before submitting.
- Keep the repository organized into notebooks, dashboard, data and report folders.
- After adding the real GitHub URL, re-export the final PDF and upload that PDF to the Coursera AI-Graded assignment.

GitHub URL: _____

Appendix — Coursera Grading Coverage

- 1. GitHub URL — placeholder remains until your own repository is available.
- 2. Completed PDF presentation — included.
- 3. Executive Summary — included.
- 4. Introduction — included.
- 5–7. API, Web Scraping and Data Wrangling — included.
- 8–10. EDA visualization, SQL and interactive analytics — included.
- 11–14. Visualization results, SQL results, Folium and Plotly Dash — included.
- 15. Predictive analysis, evaluation, confusion matrix and conclusion — included.
- Additional creativity/insights — included through the integrated workflow, appendix and explanatory analysis.